On The Relation Between The Gamma And Lambert-W Functions

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1 Definitions

The Lambert-W function, $W(x)$, constitutes the inverse of the function $f(x) = xe^x$ and thus obeys the property $f\left(f^{-1}(x)\right) = W(x)\, e^{W(x)} = x$, similar to all other inverse functions. The Lambert-W function, or product-log, as it is otherwise referred to, is used in diverse areas of mathematics such as algebra, complex analysis, and mathematical modeling.

The Gamma function, $\Gamma(x)$, may be considered the generalized factorial-function whose primary integral definition is:

$$\Gamma(x) = \int_0^{\infty} t^{x-1} e^{-t} dt$$

For positive integers n, the factorials obey the recursive property $(n+1)! = (n+1) \cdot n!$, and with the fact that $\Gamma(1) = 1$ and $\Gamma(x+1) = x\Gamma(x)$ (integration by parts), the Gamma function inductively interpolates the traditional factorials.

$$\Gamma(x+1) = \int_0^{\infty} t^x e^{-t} = \left[-t^x e^{-t}\right] - \int_0^{\infty} -xt^{x-1} e^{-t} dt = x \int_0^{\infty} t^{x-1} e^{-t} dt = x\Gamma(x)$$

In whole, the Gamma function can continuously extend the factorial notion $\Gamma(x) = (x-1)!$ to complex values, fractions, and more.

2 Development Of The Integral

Through the evaluation of the following integral containing the Lambert-W function, a relation to the Gamma function will be established.

$$\int_0^{\infty} W\left(\frac{1}{x^n}\right) dx$$

By introducing the substitution $u = \frac{1}{x^n} = x^{-n}$, where $x = \left(\frac{1}{u}\right)^{\frac{1}{n}}$ and $\frac{du}{-n\left(\frac{1}{u}\right)^{\frac{1}{n}(-n-1)}} = dx$ the integral becomes;

$$\int_{\infty}^{0} \frac{W(u)}{-n\left(\frac{1}{u}\right)^{\frac{1}{n}(-n-1)}} du = \frac{1}{n} \int_0^{\infty} \frac{W(u)}{u^{\frac{n+1}{n}}} du$$

Next, substituting $t = W(u)$, the inverse property of the Lambert-W function can be employed to extract $u = te^t$, where $du = (te^t + e^t)\, dt = e^t (t+1)\, dt$.
The bounds of the integral after the variable substitution remain the same as when t tends to either zero or infinity, the $u = te^t$ replacement does as well.

$$\frac{1}{n} \int_0^{\infty} \frac{t}{(te^t)^{\frac{n+1}{n}}} e^t (t+1)\, dt = \frac{1}{n} \int_0^{\infty} \frac{te^t (t+1)}{(t)^{\frac{n+1}{n}} (e^t)^{\frac{n+1}{n}}} dt = \frac{1}{n} \int_0^{\infty} \frac{(t+1)}{t^{\left(\frac{n+1}{n}-1\right)} e^{t\left(\frac{n+1}{n}-1\right)}} dt$$

Through the additive linearity of the integral, the Gamma function emerges;

$$\frac{1}{n}\left(\int_0^{\infty} t^{-\left(\frac{n+1}{n}-1\right)} e^{-t\left(\frac{n+1}{n}-1\right)} dt + \int_0^{\infty} t^{-\left(\frac{n+1}{n}-2\right)} e^{-t\left(\frac{n+1}{n}-1\right)} dt\right)$$

Defining the substitution $v = t\left(\frac{n+1}{n}-1\right)$ in which $\frac{dv}{\left(\frac{n+1}{n}-1\right)} = dt$ and $\frac{v}{\left(\frac{n+1}{n}-1\right)} = t$ produces;

$$\frac{1}{n}\frac{1}{\left(\frac{n+1}{n}-1\right)}\left(\left(\frac{n+1}{n}-1\right)^{\left(\frac{n+1}{n}-1\right)}\int_0^\infty v^{-\left(\frac{n+1}{n}-1\right)}e^{-v}dv + \left(\frac{n+1}{n}-1\right)^{\left(\frac{n+1}{n}-2\right)}\int_0^\infty v^{-\left(\frac{n+1}{n}-2\right)}e^{-v}dv\right)$$

$$\frac{\left(\frac{n+1}{n}-1\right)^{\left(\frac{n+1}{n}-2\right)}}{n\left(\frac{n+1}{n}-1\right)}\left(\left(\frac{n+1}{n}-1\right)\int_0^\infty v^{\left(-\frac{n+1}{n}+2-1\right)}e^{-v}dv + \int_0^\infty v^{\left(-\frac{n+1}{n}+3-1\right)}e^{-v}dv\right)$$

$$\frac{\left(\frac{n+1}{n}-1\right)^{\left(\frac{n+1}{n}-3\right)}}{n}\left(\left(\frac{n+1}{n}-1\right)\Gamma\left(-\frac{n+1}{n}+2\right)+\Gamma\left(-\frac{n+1}{n}+3\right)\right)$$

Utilizing the property $\Gamma(x+1) = x\Gamma(x)$ addressed in Section 1, the result can be further simplified to;

$$\frac{\left(\frac{n+1}{n}-1\right)^{\left(\frac{n+1}{n}-3\right)}\Gamma\left(-\frac{n+1}{n}+2\right)}{n}\left(\left(\frac{n+1}{n}-1\right)+\left(-\frac{n+1}{n}+2\right)\right) = \frac{\left(\frac{1}{n}\right)^{\left(\frac{1}{n}-2\right)}\Gamma\left(-\frac{1}{n}+1\right)}{n}$$

$$\int_0^\infty W\left(\frac{1}{x^n}\right)dx = n^{-\frac{1}{n}+1}\Gamma\left(-\frac{1}{n}+1\right) = -n^{-\frac{1}{n}}\Gamma\left(-\frac{1}{n}\right) \tag{1}$$

2.1 Manipulation

First, for conciseness, the product-log integral and resulting Gamma-function expression will be denoted as $\Xi[n]$.

$$-n^{-\frac{1}{n}}\Gamma\left(-\frac{1}{n}\right) = \Xi[n]$$

Second, through yet another manipulation that being $\xi = -\frac{1}{n}$, one can express the Gamma function as;

$$\int_0^\infty W\left(\frac{1}{x^n}\right)dx = -n^{-\frac{1}{n}}\Gamma\left(-\frac{1}{n}\right)$$

$$\int_0^\infty W\left(\frac{1}{x^{\frac{-1}{\xi}}}\right)dx = -\frac{-1}{\xi}^{-\frac{1}{\frac{-1}{\xi}}}\Gamma\left(-\frac{1}{\frac{-1}{\xi}}\right)$$

$$-(-\xi)^{\xi}\int_0^\infty W\left(x^{\frac{1}{\xi}}\right)dx = \Gamma(\xi) \tag{2}$$

Due to the potential for divergence of the improper integral, such a rearrangement holds for $0 > \xi > -1$.

3 Implications

3.1 $\Xi[2] = \sqrt{2\pi}$

For the case of $n = 2$, one finds $\int_0^\infty W\left(\frac{1}{x^2}\right)dx = 2^{-\frac{1}{2}+1}\Gamma\left(-\frac{1}{2}+1\right)$, where $\Gamma\left(\frac{1}{2}\right)$ can be evaluated by its defining integral;

$$\Gamma\left(\frac{1}{2}\right) = \int_0^\infty e^{-t}\cdot t^{\frac{1}{2}-1}dt = \int_0^\infty e^{-t}\cdot t^{\frac{-1}{2}}dt$$

By introducing the substitution $u = t^{\frac{1}{2}}$, where $t = u^2$ and $2udu = dt$, one finds half of the Gaussian integral, who's evaluation is $\frac{\sqrt{\pi}}{2}$.

$$\int_0^\infty e^{-u^2}\cdot\frac{1}{u}2udu = 2\int_0^\infty e^{-u^2}du$$

$$\int_0^\infty e^{-u^2}\cdot\frac{1}{u}2udu = 2\int_0^\infty e^{-u^2}du = \frac{2\sqrt{\pi}}{2} = \sqrt{\pi}$$

Where substitution into formula (1) yields;

$$\int_0^\infty W\left(\frac{1}{x^2}\right) dx = 2^{-\frac{1}{2}+1}\Gamma\left(-\frac{1}{2}+1\right) = 2^{\frac{1}{2}}\sqrt{\pi} = \sqrt{2\pi}$$

A standard result, yet interesting nonetheless.

3.2 Defying Ξ Convergence

Examining the integrand (Eq. 1), it is clear that the output (y-coordinate) of the implicit curve defining the Lambert-W function $W\left(\frac{1}{x^n}\right) := ye^y = \frac{1}{x^n}$ for the stipulated range of $x \geq 0$ cannot be negative, since the positive $\frac{1}{x^n}$ term would be positive only if y remains greater than 0.

Furthermore, the infinite upper bound of the improper integral defining $\Xi\left[n\right]$ permits convergence only for reciprocals of x of power $n > 1$. Both of which are observed in the graph of $\Xi\left[x\right]$ below.

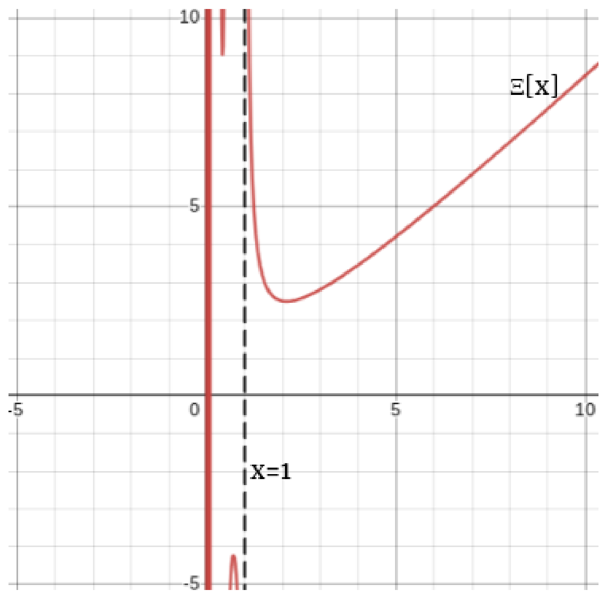

Figure 1: $\Xi\left[x\right]$ alongside asymptote $x = 1$

That stated, an inspection of values less than the vertical asymptotic threshold for convergence seem to produce an array of quite intriguing results such as $\Xi\left[\frac{2}{3}\right]$:

$$\Xi\left[\frac{2}{3}\right] = \left(\frac{2}{3}\right)^{-\frac{3}{2}+1}\Gamma\left(-\frac{3}{2}+1\right) = \sqrt{\frac{3}{2}}\cdot\Gamma\left(\frac{-1}{2}\right) = -2\sqrt{\frac{3\pi}{2}}$$

Which, by observation of the graph of $W\left(\frac{1}{x^{\frac{2}{3}}}\right)$ (Figure 2) seems nonsensical as the result insinuates a convergent value for patently non-negative function.

Another such example is of $\Xi\left[-1\right] = -\left(-1\right)^{-\frac{1}{(-1)}}\Gamma\left(-\frac{1}{(-1)}\right) = 1$, which corresponds to the integral $\int_0^\infty W\left(x\right)dx$, despite the clear unbounded nature of Lambert-W of x tends to infinity. (Figure 2)

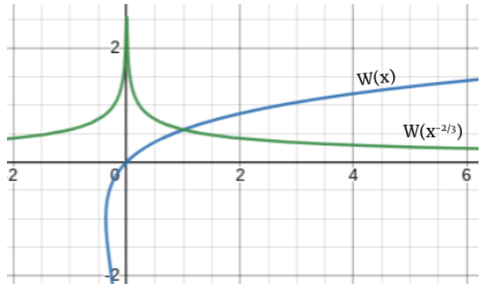

Figure 2: Graph of $W\left(\frac{1}{x^{\frac{2}{3}}}\right)$ and $W\left(x\right)$